

Comparing Schemes and Manifolds

This is just a quick extract from [1] that I wanted to post that compares schemes and smooth manifolds. It is not comprehensive, it is just the first things that came to mind. For a full discussion, see the cited notes.

1. The scheme given by $V(x) \cup V(yz) \subseteq \mathbb{A}_k^3$ has two irreducible components of different dimensions (dimension is defined rigorously in section [1, dimension]). A manifold (differential and even topological) will always be of a single dimension.
2. A manifold cannot self-intersect¹, while schemes (even affine schemes, even varieties) can: consider $V(y^2 - x^3 - x^2) \subseteq \mathbb{A}_k^2$.
3. The patches of manifold are very regular, being diffeomorphic to $\mathbb{R}^n$, however patches of a scheme may be “pathological”, for example if the ring is not Noetherian then there may be uncountably many connected components. This reflects how affine schemes represent “local algebraic information” which is more complicated than “local smooth linear information” that real manifolds work with. Some of the conditions and properties we shall introduce later will add some regularity to the components.
4. In Manifold Theory, the notion of irreducible component is trivial as only points are irreducible. From an analytic perspective, this can be thought of as the lack of separability of points by functions for schemes; part of this consequence is the existence of “large” points for schemes. More generally the correspondence between irreducible sets and non-closed points (in at least Noetherian schemes) shows that schemes have a lot more “global” information on its codimension subsets as compared to real smooth manifolds.
5. The topology of $\mathbb{R}^n$ (and hence $\mathbb{C}^n$) is the product of open sets of $\mathbb{R}$, namely

$$\prod_i^n (a_i, b_i) \quad a_i, b_i \in \mathbb{R}, \quad a_i < b_i$$

forms a topological basis for $\mathbb{R}^n$. This is far from the case for $\mathbb{A}_k^n$: its closed sets would always be finite! However, there will still be a meaningful way to create the topology on $\mathbb{A}_k^n$ via $\mathbb{A}_k^1$ by taking the *fibred product*

6. If a scheme X is irreducible, all its open sets are dense. Many of the schemes we shall work with will be the union of irreducible subschemes, where for each irreducible subscheme all affine open subsets are dense in the irreducible subscheme; having dense open subsets is far from being the case for most manifolds.
7. More generally, the behaviour of schemes matches more closely with the behaviour of complex geometry (we shall draw such analogies often, for example *Hartog’s Extension Lemma* ([5, chapter 8]) shall have parallels with normal schemes, the existence of certain holomorphic/meromorphic functions shall tie into divisor theory, compact complex Riemann surfaces shall correspond to projective curves, and so forth).

¹Naturally, a submersion of a manifold may have intersections, however (naturally) the immersion is not a manifold

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8. Any smooth real manifold M can be embedded into $\mathbb{R}^N$ for appropriate N . For classical affine (resp. projective) varieties of dimension n , they essentially by definition embed into $\mathbb{A}_k^n$ (resp. $\mathbb{P}_k^n$). For complex manifolds, it is *not* the case they all embed in $\mathbb{C}^{N^2}$, and no compact complex scheme embeds into $\mathbb{C}^N$. This is because complex manifolds have much more rigidity that $\mathbb{C}^n$ lacks [2]³. Schemes follow in this complexity.

By the Kodaira Embedding Theorem [2], a “positive Khähler manifold” can embed into projective space. Proper schemes (the equivalent of compact complex manifolds) with an *ample line bundle* (the equivalent of positive Khähler structure) shall embed into projective space. In fact, this means that positive Khähler manifolds are isomorphic to a proper scheme with an ample bundle!. This is the closest to an embedding theorem that is widely used and will be proven in section [1, Maps to Projective Schemes].

For a closed immersion $X \hookrightarrow \mathbb{A}_R^n$ so that $X \cong V \subseteq \mathbb{A}_R^n$ where V is closed, this means that X is topologically sub-variety, though it may have a more complicated sheaf structure (for more details see section [1, Varieties]).

For an open embedding $X \hookrightarrow \mathbb{A}_R^n$ so that $X \cong U \subseteq \mathbb{A}_R^n$ where U is open, this means that X is a *quasi-affine subscheme*, which by definition is an open subscheme of affine space. Their characterization is more complicated: the map $X \rightarrow \text{Spec}(\mathcal{O}_X(X))$ must be an open embedding and, as will be discussed in section [1, Quasi-Coherent Sheaves], all quasi-coherent sheaves of $\mathcal{O}_X$ -modules must be generated by global sections. To check that each $\mathcal{O}_X$ -module is generated by global sections is difficult, there is a simpler characterization by the Goodman-Hartshorne Theorem [3], but it is still a complicated characterization showing the greater complexities that arise when working with non-proper schemes.

More generally, there shall be other interesting spaces to embed certain types of schemes into (ex. into toric varieties, Grassmannians, flag varieties, etc.) which shall have different purposes depending on what Moduli problem is being studied.

9. Schemes will be able to have “double points”, recall the line with two origin is a popular counter for a manifold as it is not Hausdorff at the origin. These shall be important as many moduli spaces will be non-separated schemes.
10. A scheme remembers multiplicity. For example, we can define a lot of functions on a single point, however scheme functions (i.e. regular functions) can hold multiplicity information (ex. $\text{Spec } k[x]/(x)^2$). A smooth manifold $(M, \mathcal{A})$ with maximal atlas does not hold this information.
11. There can be multiple structure sheaves on the same topological space defining different schemes (think of the different fields that can form the sheaf over a singleton). Manifolds of dimension ≤ 3 have unique differential structures up to diffeomorphism. In fact, in section [1, Fibered Products] we shall introduce the process of *base change* from which there shall be ways to change the structure sheaf without changing the underlying space.
12. Though there wasn’t an atlas defined for schemes, schemes can be constructed via gluing affine just like an atlas on a manifold.
13. the natural additional structure on a smooth manifold is a *tangent bundle* with transition mappings of the form $D(\mathbf{y} \circ \mathbf{x}^{-1})$, namely they are not just smooth transitions between open sets and fibers, but need to respect a $\text{GL}_n(\mathbb{R})$ -action given by the derivation D . We shall define the equivalent for schemes in section [1, Differentials], but more care must be done as schemes are not “locally linear” everywhere in the way that smooth manifolds must be. Much of the intuition of Poincaré’s duality (see [4, section 4.3]) will influence the generalization.

²If it does embed, such complex manifold are called *stein manifolds* and they share many properties of quasi-projective varieties

³consider that there is a unique complex structure on the complex torus for each complex number

14. Schemes shall be able to better remember different levels of *local* behaviour. For example, a germ is not a smooth manifold on a point, but the equivalent for a scheme shall be a scheme, known as a *local scheme*. This is because commutative rings represent this information, and (affine) schemes are the geometric representation of commutative rings.
15. Scheme functions are more “rigid”; while on a manifold we may have partitions of unity due to the flexibility of euclidean topology, no exact equivalent can be defined for schemes. This rigidity will also add structure and make it easier to defined some concepts (ex. closed subschemes, introduced in section [1, Open and Closed Subschemes], shall be a rather “natural” concept as compared to the work that goes into ensuring closed submanifolds are locally isomorphic to $\mathbb{R}^k$).
As smooth functions on each $U \subseteq M$ from a sheaf, sheaves that admit a “partition of unity” will be called *fine sheaves*. A weaker notion of *soft sheaf* will be introduced in section [1, Sheaf Cohomology] that will be the closest equivalent to having the partition of unity property.
16. Schemes unify “arithmetic” and “geometry” in a way manifold theory is poorly equipped for. As an example, there is no equivalent of $\text{Spec } \mathbb{Z}$, $\text{Spec } \mathcal{O}_K$ or $\text{Spec } \mathbb{Z}[t]$ where $\mathcal{O}_K$ is a ring of integers. In this way, scheme theory also allows for a new type of object that is not tractable in differential geometry. Often, to deal with more “geometric” schemes, one works with $\bar{k}$ -schemes where $\bar{k}$ denotes the algebraic closure of a characteristic 0 field. In fact, when defining Z -points in section ?? where Z is a scheme, if $Z = \text{Spec } \bar{k}$ then such points shall be called *geometric points*. Working over a non-algebraically closed field often implies the schemes contains some arithmetic information.
17. Schemes have “fat” points. Such points are necessary for examples such as $k[x, y] \rightarrow k(x)[y]$ whose Spec dual will map closed points to non-closed points (and we shall later see the image is *dense*).
18. Though the above show how schemes differ from manifolds geometrically, many of the usual geometric tools are applicable in the study of schemes. For example, $\mathbb{C}[x, y](x^2 - y^3)$ is not a normal ring. This can be hard to see, but viewing the Spec of it, the fact it is non-smooth (in the euclidean topology, as a closed subset of $\mathbb{C}^2$) at $(0, 0)$ is in fact exactly why it is not normal. This type of geometric shortcut can be incredibly powerful as we can now study the smoothness (more particularly the normalness, meaning smoothness in codimension 1) to determine the integral closure of $k[x, y]$.

Though this is a long list of mostly differences, very often in practice (especially in the setting of k -schemes or when studying curves, surfaces, moduli spaces, etc.) we work with schemes that most of the time are proper (the correct generalization of compact manifolds), and have many methods to “normalize” singularities (ex. eliminate intersections, cusps, etc) to make these scheme regular⁴. In fact, as we shall see in section [1, gaga], a result known as *GAGA* shows when proper schemes with appropriate coherent sheaves are equivalent to a compact complex manifolds. Naturally, arithmetic schemes will not be part of this manifold-scheme correspondence.

References

- [1] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Geometry*. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algGeo.pdf.

⁴Though as we’ll see, there is no universal way of doing this with the exception of some low-dimension that have been completely proved, and sometimes this singularity information is important.

- [2] Nathanael Chwojko-Srawley. *Everything You Need To Know About Hodge Theory*. Vol. 1. URL: https://nathanaelsrawley.com/pdf-viewer?src=.%2F.%2Fassets%2Fpdfs%2Fnotes%2FEYNTKA_hodge.pdf&name=A%20little%20%20Hodge%20Theory&source=notes.
- [3] Robin Hartshorne. *Ample subvarieties of algebraic varieties*. eng. [Nachdr. der Ausg.] 1970. Lecture notes in mathematics Series: Tata Institute of Fundamental Research, Bombay. Springer, 2009. ISBN: 978-3-540-05184-8.
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- [5] Nathanael Chwojko-Srawley. *Everything You Need To Know About Complex Analysis*. 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_complex_analysis.pdf.